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# 1 Introduction

# 2 Networks Fundamentals

## 2.1 Types of networks in terms of geographical scope

we have many types of networks in terms of geographical scope, such as:

- Personal Area Network (PAN)
- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

let's briefly explain each of these types:

### 2.1.1 Personal Area Network (PAN)

This type of network is private, and its range does not exceed 10 meters. It is used to connect two devices such as computers, fax machines, or printers. It most commonly uses Bluetooth technology, meaning that the connection is wireless and operates through radio waves

### 2.1.2 Local Area Network (LAN)

A Local Area Network (LAN) connects multiple computers and devices within a small geographic area such as a building, or company. It is one of the most common network types, typically owned and managed by the organization, and often uses Ethernet technology for communication.

### 2.1.3 Metropolitan Area Network

A Metropolitan Area Network (MAN) connects multiple Local Area Networks (LANs) across a larger geographic area such as a city or metropolitan region. It is commonly used by large organizations, universities, government institutions, or service providers to link multiple buildings or branches within the same city. MAN networks often rely on high-speed connections such as fiber optics and Ethernet technology to ensure fast and reliable communication.

#### 2.1.4 Wide Area Network (WAN)

A Wide Area Network (WAN) is an open and widely used network type with no geographical limitations. Its main function is to connect countries, distant cities, local networks, and company branches together. It is considered one of the largest types of networks in the world.

also we have ather types of networks in terms of their purpose, such as:

- Storage Area Network (SAN)
- Virtual Private Network (VPN)
- Wireless Local Area Network (WLAN)

#### 2.1.5 Network infrastructure

It consists of the physical components. These components are installed within the network and enable the devices to connect with one another and exchange data among themselves. These components include cables, network points, and network devices such as routers, switches, and servers.

### 2.2 Network Architectures

#### 2.2.1 Peer – to – Peer Networks

A Peer-to-Peer (P2P) network allows users to share files, printers without a centralized server or control unit. Any user can connect to the network, and each user manages and installs their own software. The network has a limited capacity, typically supporting up to 20 computers, and is commonly referred to as a Workgroup. Additionally, there is no centralized storage system, as each user maintains their own local storage.

#### 2.2.2 Client – Server Networks

Client–Server architecture is a network model in which one or more central servers provide services and resources to multiple client devices. The server manages data, security, storage, and network control, while clients send requests and receive responses. This model offers centralized management, improved security, scalable performance, and organized resource sharing, making it suitable for medium to large networks such as businesses, universities, and institutions

## 2.3 Physical Network Topologies

we have many types of physical network topologies, used to network design and structure, such as:

- Bus Topology
- Star Topology
- Ring Topology
- Mesh Topology
- Tree Topology
- Hybrid Topology

## 2.4 Physical Media

we have 3 common cables types :

- Cable Coaxial
- Twisted Pair Cable
- Fiber Optic Cable

### 2.4.1 Cable Coaxial

A copper cable with a central conductor surrounded by insulation and a metallic shield.
**Speed:** Medium (up to hundreds of Mbps).
**Characteristics:** Good shielding, resistant to interference, durable.
**Usage:** Cable TV, CCTV, broadband internet (older systems).

### 2.4.2 Twisted Pair Cable

wo insulated copper wires twisted together to reduce interference. **Speed:** Medium to High (up to 10 Gbps with Cat6a/Cat7).
**Characteristics:** Low cost, easy installation, flexible.
**Usage:** Ethernet networks (LAN), telephone lines.
**Types:**

- STP (Shielded Twisted Pair)
- UTP (UnShielded Twisted Pair)

**Connection Types**

- Straight-Through Cable: Used to connect different devices (e.g., computer to switch).
- Crossover Cable: Used to connect similar devices (e.g., computer to computer

because the wires are crossed in the cable, allowing for direct communication between devices without a switch or hub. TX of Device A must connect to RX of Device B and vice versa.

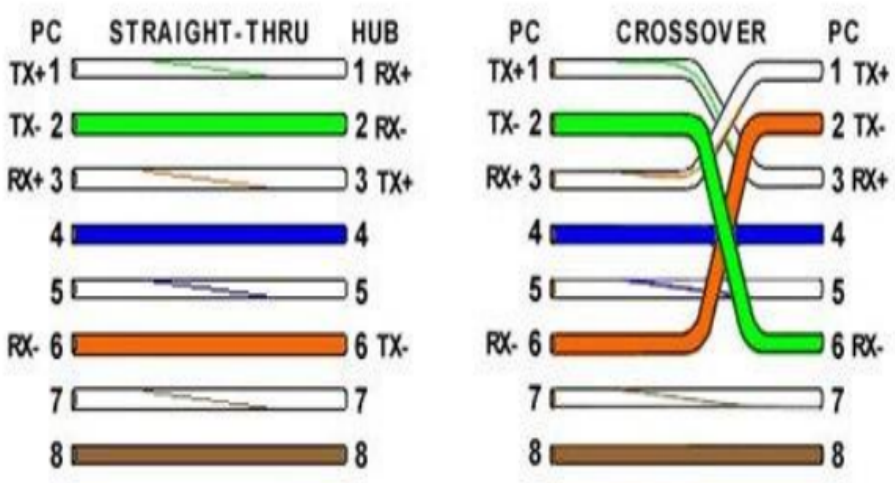


Figure 1: Straight-Through vs Crossover Cable

### 2.4.3 Fiber Optic Cable

A cable made of glass or plastic fibers that transmit data as light signals.
**Speed:** Very High (up to 100 Gbps and beyond).
**Characteristics:** Immune to electromagnetic interference, long-distance transmission, high bandwidth.
**Usage:** Long-distance communication, backbone networks, data centers.

## 2.5 Network Devices

we have a lot of network devices, and each one has own os and function in specific network layer , such as:

### 2.5.1 Network Card - NIC

is a hardware device that allows a computer to connect to a network. It operates at the Physical and Data Link layers, converts data into signals for transmission, receives incoming data, and provides a unique MAC address for network identification.

### 2.5.2 Hub

A hub is a basic networking device that connects multiple computers in a local area network (LAN). It operates at the Physical layer (Layer 1) of the OSI model and broadcasts incoming data to all connected devices without filtering or addressing.

### 2.5.3 Switch

A switch is a networking device that connects multiple devices within a local area network (LAN) and forwards data only to the intended destination device. It operates at the Data Link layer (Layer 2) of the OSI model and uses MAC addresses to make forwarding decisions

### 2.5.4 Router

A router is a networking device that connects different networks together and forwards data packets between them. It operates at the Network layer (Layer 3) of the OSI model and uses IP addresses to determine the best path for data transmission.

### 2.5.5 Pc - Personal Computer

A PC is an end device, so it sends and receives data and uses all OSI layers — from the Physical layer (signals and cables) up to the Application layer (user programs like browsers and email).

### 2.5.6 Access Point

Access Point (AP) is a networking device that allows wireless devices to connect to a wired local area network (LAN) using Wi-Fi.

### 2.5.7 Server

A Server is a computer system that provides services, resources, or data to other computers (clients) over a network. Operates at: All 7 OSI layers (like a PC), because it is an end device.

### 2.5.8 Modem

A Modem (Modulator-Demodulator) is a device that converts digital signals into analog signals and vice versa for transmission over communication lines.

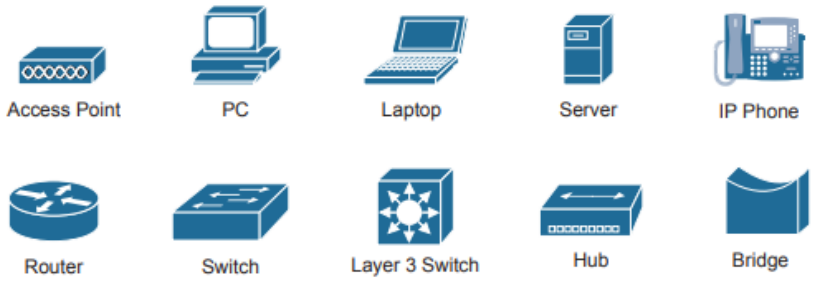


Figure 2: Icons devices

## 2.6 Networks Architecture Design

we used topologies and cables and devices to design the network architecture, and we have many types of network architecture design, such as:

### 2.6.1 Flat Network Design

All devices connect to a single layer of switches (no hierarchy)
**Advantages:** Simple, easy to set up, low cost.
used to be common in small networks but can lead to performance issues and security risks as the network grows.

### 2.6.2 Three-Tier Architecture (3-Layer Design)

is a hierarchical network design that consists of three layers: Core, Distribution, and Access.
**Advantages:** Scalable, high performance, better security, easier management
**Usage:** Large enterprise networks, data centers. but is more complex and expensive to implement than flat design, making it less suitable for small networks.

#### 2.6.3 Collapsed Core (Two-Tier Design)

is a simplified version of the three-tier architecture that combines the Core and Distribution layers into a single layer.
**Advantages:** Reduced complexity, lower cost, easier management.
**Usage:** Medium-sized networks, small enterprises. However, it may not provide the same level of performance and scalability as a full three-tier design, making it less suitable for large networks with high traffic demands.

#### 2.6.4 Spine-Leaf Architecture

is a modern network design commonly used in data centers that consists of two layers: Spine and Leaf.
**Advantages:** High performance, low latency, scalable, better east-west traffic handling.
**Usage:** Data centers, cloud environments. However, it can be more complex and costly to implement than traditional architectures, making it less suitable for small networks or organizations with limited resources.

### 2.7 Protocols

### 2.8 Open Systems Interconnection

### 2.9 TCP/IP Protocol Suite